Pulmonary Disorders I

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Learning Objectives

1. Evaluate the role of pharmacologic management options in acute respiratory distress syndrome.
2. Review clinical practice guidelines pertaining to acute respiratory distress syndrome.
3. Recommend an evidence-based approach for non-pharmacologic therapy in managing critically ill patients with acute respiratory distress syndrome.
4. Evaluate key variables and commonly used modes for treatment with mechanical ventilation.
5. Assess appropriateness of drug therapy for endotracheal intubation, including agents for premedication, induction, and neuromuscular blockade.

Abbreviations in This Chapter

AC/VC	Assist control/volume control
ARDS	Acute respiratory distress syndrome
COVID-19	Coronavirus disease 2019
CVP	Central venous pressure
ECMO	Extracorporeal membrane oxygenation
ICU	Intensive care unit
MAP	Mean arterial pressure
MV	Mechanical ventilation
NMBA	Neuromuscular blocking agent
PEEP	Positive end-expiratory pressure
PS	Pressure support
RSI	Rapid sequence intubation
SIMV	Synchronized intermittent mandatory ventilation

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. Which best describes the category of acute respiratory distress syndrome (ARDS) that most benefits from prone positioning and cisatracurium administration?
   A. Acute lung injury.
   B. Moderate to severe.
   C. Mild to moderate.
   D. Mild.

2. Several clinical trials have been conducted in moderate to severe ARDS. Which of the following partial pressure of arterial oxygen (Pa_{O_2})/ fraction of inspired oxygen (F_{IO_2}) parameters best describes this ARDS subpopulation?
   A. 100–200 mm Hg.
   B. Greater than 150 mm Hg.
   C. Less than 150 mm Hg.
   D. 50–150 mm Hg.

3. Which of the following parameters best describes "low tidal volumes" for invasive mechanical ventilation in ARDS?
   A. 6 mL/kg of total body weight.
   B. 4 mL/kg of ideal body weight.
   C. 7 mL/kg of total body weight.
   D. 5 mL/kg of adjusted body weight.

4. A 65-year-old man presents to the emergency department with severe shortness of breath, tachypnea, altered mental status, and diaphoresis. A chest radiograph reveals diffuse, bilateral opacities. His vital signs are as follows: blood pressure 94/54 mm Hg, respiratory rate 26 breaths/minute, heart rate 120 beats/minute, pain score 2/10, and temperature 105.8°F (41.0°C). The patient's wife states that his symptoms began about 2 days ago and gradually worsened over the past day. The patient is transferred to the ICU, where he is intubated. His arterial blood gas values are pH 7.30, partial pressure of arterial carbon dioxide (Pa_{CO_2}) 50 mm Hg, Pa_{O_2} 50 mm Hg, and oxygen saturation (Sa_{O_2}) 85% while receiving F_{IO_2} 100%. Which is the best therapy plan for the next 24 hours?
   A. Empiric antibiotic therapy, intravenous fluid resuscitation, and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a Richmond Agitation-Sedation Scale (RASS) score of –4; and prone positioning.
   B. Empiric antibiotic therapy, intravenous fluid resuscitation, and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a RASS score of –5 and cisatracurium administration to limit plateau pressures; and prone positioning.

C. Empiric antibiotic therapy, aggressive diuresis (goal central venous pressure less than 4 mm Hg), and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a RASS score of −4; and prone positioning.
D. Empiric antibiotic therapy, intravenous fluid resuscitation, and vasopressors for shock; low tidal volume (4–8 mL/kg) ventilation strategy; deep sedation to achieve a RASS score of −4; and supine positioning.

5. P.D. is a 67-year-old woman (height 160 cm; total body weight 68 kg) admitted to the ICU with ARDS pneumonia. Her mechanical ventilation parameters are as follows: tidal volume 280 mL, respiratory rate 20 breaths/minute, positive end-expiratory pressure (PEEP) 12 cm H_2O, and F_{IO_2} 80% with plateau pressure of 22 mm Hg. Her mean arterial pressure (MAP) is currently 68 mm Hg with norepinephrine discontinued 4 hours ago and CVP ranging between 8 and 14 cm H_2O over the last 2 hours. Also, her total urine output over the past 24 hours is 1150 mL. The ICU team is discussing fluid management using the FACTT Lite protocol. Which of the following options is the most appropriate strategy?
   A. Administer furosemide 40 mg IV once and re-evaluate in 4 hours.
   B. Administer furosemide 1mg/hour continuous infusion and re-evaluate in 8 hours.
   C. Administer 0.9% sodium chloride 500 mL over 1 hour and re-evaluate in 4 hours.
   D. Monitor only and re-evaluate in 8 hours.

6. Which of the following statements most appropriately describes published clinical trials of corticosteroids in ARDS?
   A. Dexamethasone demonstrated improved clinical outcomes among ARDS patients on lung-protective strategies.
   B. Dexamethasone initiated after 7 days of ARDS onset was associated with worse clinical outcomes.
   C. Methylprednisolone demonstrated improved clinical outcomes among ARDS patients on lung-protective strategies.
   D. Methylprednisolone initiated after 7 days of ARDS onset was associated with worse clinical outcomes.

7. Which order of medication administration would be most appropriate for a 34-year-old woman with no significant medical history receiving rapid sequence intubation (RSI)?
   A. Rocuronium, etomidate, midazolam.
   B. Fentanyl, succinylcholine, propofol.
   C. Atropine, rocuronium, etomidate.
   D. Fentanyl, etomidate, succinylcholine.

8. A 78-year-old man (weight 70 kg) presents to the ICU after being intubated for acute respiratory failure. His ventilator settings are as follows: assist control/volume control mode, tidal volume 700 mL (10 mL/kg), respiratory rate 20 breaths/minute, F_{IO_2} 50%, PEEP 5 cm H_2O, and pressure support 10 cm H_2O. The first arterial blood gas values are pH 7.25, Pa_{CO_2} 65 mm Hg, HCO_3 26 mEq/L, Pa_{O_2} 65 mm Hg, and Sa_{O_2} 90%. His MAP is 72 mm Hg without requiring vasoactive support, with a serum creatinine of 1.2 mg/dL and urinary output of 350 mL over the past 6 hours. Which approach would be best for this patient's initial treatment?
   A. High tidal volume mechanical ventilation (MV) and conservative fluid management.
   B. Low tidal volume MV and conservative fluid management.
   C. High tidal volume MV and liberal fluid management.
   D. Low tidal volume MV and liberal fluid management.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
    a. Task A: 1, 2, 4, 6
2. Domain 2: Therapeutics and Patient Management
    a. Task A: 1, 3, 6, 7
    b. Task B: 1, 13
    b. Task C: 1
3. Domain 3: Personal Practice
    a. Task B: 1

I. ACUTE RESPIRATORY DISTRESS SYNDROME

A. Definition and Epidemiology
   1. First described in 1967; several definitions have evolved over the years
   2. Berlin ARDS definition was established in 2012 to address issues with previous definitions (Table 1) (JAMA 2012;307:2526-33).

Table 1. Berlin ARDS Definition

Variable	Criteria
Timing	Onset within 1 wk of a known clinical insult or new or worsening respiratory symptoms
Chest imaging	Bilateral opacities not fully explained by effusions, lobar/lung collapse, or nodules
Origin of edema	Respiratory failure not fully explained by cardiac failure or fluid overload Need objective assessment (ECHO) to exclude hydrostatic edema if no risk factors present
Oxygenation	
Mild	200 mm Hg < Pa_{O_2}/F_{IO_2} ≤ 300 mm Hg with PEEP or CPAP ≥ 5 cm H_2O
Moderate	100 mm Hg < Pa_{O_2}/F_{IO_2} ≤ 200 mm Hg with PEEP ≥ 5 cm H_2O
Severe	Pa_{O_2}/F_{IO_2} ≤ 100 mm Hg with PEEP ≥ 5 cm H_2O

ARDS = acute respiratory distress syndrome; CPAP = continuous positive airway pressure; ECHO = echocardiogram; F_{IO_2} = fraction of inspired oxygen; Pa_{O_2} = partial pressure of oxygen; PEEP = positive end-expiratory pressure..

   3. ARDS represents 10.4% of all ICU admissions and 23.4% of patients requiring mechanical ventilation (MV) (JAMA 2016;315:788-800).
      a. Mild ARDS accounts for 30%, moderate for 46.6%, and severe for 23.4%.
      b. Hospital mortality is 34.9% for mild ARDS, 40.3% for moderate ARDS, and 46.1% for severe ARDS.
      c. Multisystem organ failure is the leading cause of death in patients with ARDS, with the number of extrapulmonary organ failures correlating with an incremental increase in mortality (Intensive Care Med 2011;37:1932-41).

B. Cause and Pathophysiology
   1. Direct and indirect causes of lung injury
      a. Direct: Pneumonia, aspiration, trauma
      b. Indirect: Sepsis, transfusion injury, pancreatitis, burn injury, trauma
   2. Pathogenesis
      a. The hallmark clinical manifestation of ARDS is hypoxemia from alveolar collapse and edema.
      b. The exudative phase is characterized by increased permeability from the alveolar epithelium and capillary endothelial complex damage, leading to diffuse alveolar edema with fluid and cellular debris (e.g., neutrophils, cytokines, platelets). In addition, type II cells responsible for surfactant production are damaged. The peak incidence of this phase typically occurs within the first week after the initial insult.
      c. The fibroproliferative phase marks either recovery or progression of ARDS. Patients may partly or fully recover pulmonary function from drainage of the alveolar fluid, type II cellular repair, and improvement in the integrity of the endothelium-epithelium complex. However, patients whose conditions are progressively worsening may develop significant alveolar, interstitial, and capillary fibrosis. This phase typically manifests later in the course of ARDS (i.e., more than 7 days after the initial insult).

C. Management Strategies
   1. MV
      a. Lung-protective strategies using low tidal volume ventilation are considered the standard of care and cornerstone in managing ARDS.
      b. The landmark multicenter trial by the Acute Respiratory Distress Syndrome Network (ARDSNet) showed a survival benefit over conventional ventilation using higher tidal volumes (12 mL/kg) (N Engl J Med 2000;342:1301-8). Clinical practice guidelines recommend limiting tidal volumes (4–8 mL/kg of predicted body weight) (Am J Respir Crit Care Med 2017;195:1253-63; Intensive Care Med 2023;49:727-759) and inspiratory pressures (plateau pressure less than 30 cm H_2O) (Am Respir Crit Care Med 2017;195:1253-63). Although optimizing higher PEEP values may be beneficial (Am J Respir Crit Care Med 2017;195:1253-63), individualizing PEEP in ARDS remains unclear (Intensive Care Med 2023;49:727-59). Although predicted body weight and ideal body weight are not identical, clinicians often apply ideal body weight for tidal volumes for easy of use and familiarity.
   2. Prone positioning
      a. Prone over supine positioning improves gas exchange and may reduce ventilator-induced lung injury. Compression atelectasis attributed to the weight of the heart, ventral lungs, and abdominal viscera may be exacerbated in the supine position in patients with ARDS. The gravitational effects by placing a patient in the prone position improve ventilation/perfusion matching and end-expiratory lung volume by more homogeneous tidal volume delivery (Chest 2017;151:215-24; Am J Respir Crit Care Med 2017;195:1253-63).
      b. The PROSEVA study group represents the most recently published randomized clinical trial comparing prone and supine positioning in ARDS. In this trial, prone positioning for at least 16 hours/day compared with supine positioning in early ARDS decreased 28-day (adjusted hazard ratio [HR] 0.42; 95% confidence interval [CI], 0.26–0.66) and 90-day (adjusted HR 0.48; 95% CI, 0.32–0.72) mortality (N Engl J Med 2013;368:2159-68). Significantly more ventilator-free days at days 28 and 90 were observed with proning. In addition, 90-day extubation rates were significantly higher in patients in the prone position.
      c. Several meta-analyses have suggested an overall mortality benefit associated with prone positioning over supine in ARDS, though this finding was inconsistent (Intensive Care Med 2014;40:332-41; Crit Care 2014;18:R109; Crit Care Med 2014;42:1252-62; Ann Am Thorac Soc 2017;14(suppl 4):S280-S8; J Thorac Dis 2015;7:356-67; Am J Respir Crit Care Med 2021;203:1366-77). However, most meta-analyses have shown a survival benefit with proning in patients with moderate to severe ARDS (i.e., Pa_{O_2}/F_{IO_2} less than 150 mm Hg) compared with mild hypoxemia (i.e., Pa_{O_2}/F_{IO_2} greater than 200 mm Hg) (Crit Care 2014;18:R109; Crit Care Med 2014;42:1252-62; Ann Am Thorac Soc 2017;14(suppl 4):(S280-S8). In addition, mortality was reduced in patients with ARDS with prone positioning and concurrent lung-protective MV strategies compared with pronation and high tidal volume ventilation (Intensive Care Med 2014;40:332-41; Crit Care Med 2014;42:1252-62; J Thorac Dis 2015;7:356-67; Am J Respir Crit Care Med 2021;203:1366-77). Prone positioning in moderate and severe ARDS has been associated with the greatest reduction in mortality compared with other strategies, including ECMO (Am J Respir Crit Care Med 2021;203:1366-77).
      d. Implementation of prone positioning requires robust planning and resources (J Pharm Pract 2019;32:347-60). Contraindicated criteria of use should be established to mitigate any adverse events (e.g., spinal cord injury, elevated intracranial pressure [ICP]). Health care providers should be trained on appropriate movement of patients between supine and prone positioning to ensure securing of central lines as well as the endotracheal tube. Institutions should also consider the need for manual maneuvering of patients compared with specialized rotating beds. One important patient safety consideration is rapid health care staff access to patients in specialized beds during medical emergencies (e.g., cardiopulmonary arrest, self-extubation).

e. Recently published clinical practice guidelines strongly recommend prone positioning in moderate to severe ARDS (Table 2) (Am J Respir Crit Care Med 2017;195:1253-63; BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69; Intensive Care Med 2023;49:727-59). Despite overwhelming support for this therapy in ARDS, several important questions remain, including the optimal duration and timing of use pertaining to ARDS onset. Patients with ARDS should remain in the prone position for more than 12 hours each day, with consideration for longer daily durations (i.e., at least 16 hours per day). Prone positioning therapy should be considered early in the course of ICU admission in patients with moderate to severe ARDS.

Table 2. Clinical Practice Guideline Recommendations for Prone Positioning in ARDS

Source	Summary	GRADE Recommendation
Am J Respir Crit Care Med 2017;195:1253-63	Suggest prone positioning > 12 hr/day	Strong recommendation; moderate-high level of evidence
BMJ Open Respir Res 2019;6:e000420	• NOT recommended for all levels of ARDS severity • Recommended for at least 12 hr/day in moderate to severe ARDS (i.e., Pa_{O_2}/F_{IO_2} < 150 mm Hg)	Strongly in favor
Ann Intensive Care 2019;9:69	Should be used in moderate to severe ARDS (Pa_{O_2}/F_{IO_2} < 150 mm Hg) for at least 16 consecutive hr	High level of evidence
Intensive Care Med 2023;49:727-59	Recommend using early after intubation in moderate to severe ARDS (Pa_{O_2}/F_{IO_2} < 150 mm Hg for at least 16 consecutive hours)	Strong recommendation; high level of evidence

3. Extracorporeal membrane oxygenation (ECMO)
   a. The CESAR trial showed that treatment at an ECMO referral center improved 6-month survival (Lancet 2009;374:1351-63).
      i. ARDS with a Murray score (i.e., acute lung injury score on the basis of chest radiography, hypoxemia score, peak end-expiratory pressure, and static compliance) of 3 or greater, or hypercapnia and a pH less than 7.20
      ii. Not all patients randomized to the ECMO referral center received ECMO (22 of 90 patients in the experimental group did not receive ECMO).
      iii. Potential for confounding because of experiential bias – Clinicians at an ECMO treatment center may have more experience in treating ARDS.
   b. The Respiratory ECMO Survival Prediction (RESP) score was developed from a registry of patients receiving ECMO (Am J Respir Crit Care Med 2014;189:1374-82).
      i. The RESP score predicts patient survival after ECMO initiation.
      ii. Does not help quantify the decision of whether to initiate ECMO or determine whether ECMO treatment will improve survival in a specific patient
   c. In the EOLIA trial, early use of ECMO compared with conventional management in severe ARDS (e.g., Pa_{O_2}/F_{IO_2} less than 80 mm Hg) was not associated with reduced 60-day mortality rates (relative risk [RR] 0.76; 95% CI, 0.55–1.04, p=0.09) (N Engl J Med 2018;378:1965-75).
      i. This trial was prematurely discontinued because of predefined futility criteria, which resulted in not achieving power.
      ii. The high crossover rate of 28% of controls into the ECMO group may have introduced bias.

d. The 2019 clinical practice guidelines state that ECMO should be considered in patients with severe ARDS or more specifically with Pa_{O_2}/F_{IO_2} ratios less than 80 mm Hg and/or in patients with elevated plateau pressures despite optimizing PEEP, neuromuscular blocking agents (NMBAs), and prone positioning (BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69). The 2023 clinical practice guidelines recommend ECMO use in severe ARDS (as defined in EOLIA trial eligibility criteria) at an established ECMO center (strong recommendation; moderate level of evidence in favor) (Intensive Care Med 2023;49:727-59).

e. Concomitant veno-venous ECMO and prone positioning
   i. A meta-analysis (n=13 studies including only 1 randomized clinical trial) evaluated the clinical impact of prone positioning with veno-venous ECMO in ARDS compared with veno-venous ECMO alone (Intensive Care Med 2022;48:270-80).
   ii. Pooled data showed combination veno-venous ECMO and prone positioning 28-day survival was significantly improved (n=10 studies; RR 1.31; 95% CI, 1.21–1.41). Also, combination therapy was also associated with significantly improved ICU and in-hospital survival as well as overall 60- and 90-day survival.
   iii. A priori subgroup analyses found improved 28-day survival with concomitant prone positioning and veno-venous ECMO in both patient cohorts with COVID-19 ARDS patients (n=6 studies; RR 1.32; 95% CI, 1.15–1.50) and ARDS patients without COVID-19 (n=4 studies; RR 1.30; 95% CI, 1.19–1.43).
   iv. Pooled data also showed significant improvements in ventilator-free days at day 28 (n=9 studies; RR –1.29; 95% CI, –2.39 to –0.19). with prone positioning and veno-venous ECMO over ECMO alone in ARDS patients
   v. No major complications were reported with either study group.

4. Fluid management
   a. The FACTT trial compared optimal fluid management strategies – conservative (CVP less than 4 mm Hg) and liberal (CVP 10–14 mm Hg) – in patients with ARDS and hemodynamic stability (not requiring vasopressors or MAP greater than 60 mm Hg) (N Engl J Med 2006;354:2564-75). Diuretics were withheld in patients with shock but were administered according to study protocol once patients had established hemodynamic stability (discontinuation of vasopressors or MAP greater than 60 mm Hg). Although 60-day mortality did not differ (p=0.30), the conservative compared with the liberal strategy was associated with increased ventilator-free days (14.6 ± 0.5 vs. 12.1 ± 0.5, $p<0.001$) and ICU-free days (28 days) (13.4 ± 0.4 vs. 11.2 ± 0.4, $p<0.001$).
   b. Fluid strategies were compared (conservative, liberal, and simplified conservative) in a retrospective comparison among protocols pertaining to ARDS (Crit Care Med 2015;43:288-295). The FACTT Lite protocol provides fluid management recommendations pertaining to the administration of furosemide or fluids as well as monitoring without intervention-based CVP, MAP, urinary output, and pulmonary artery occlusion pressure (optional) (Table 3). No significant differences were found between the FACTT Lite and the FACTT conservative strategies for ventilator-free days (14.9 vs. 14.6, respectively; p=0.61), ICU-free days (14.4 vs. 13.4, respectively; p=0.054), or death at 60 days (22% vs. 25%, respectively; p=0.15). Compared with FACTT liberal, the FACTT Lite approach had improved outcomes (ventilator-free and ICU days and 60-day mortality).
   c. A meta-analysis evaluated the safety and efficacy of conservative versus liberal fluid strategies in ICU patients with either ARDS or sepsis (Intensive Care Med 2017;43:155-70). No differences in mortality were found between study groups in the ARDS subgroup (n=5 studies; RR 0.91; 95% CI, 0.77–1.07) or the mixed ARDS and sepsis subgroup (n=2 studies; RR 0.95; 95% CI, 0.80–1.14).

Table 3. FACTT Lite Simplified Conservative Fluid Management Approach[a]

CVP (Recommended)	Pulmonary Artery Occlusion Pressure (optional)	MAP ≥ 60 mm Hg AND No Vasoactive Support ≥ 12 Hr	
		Urinary Output < 0.5 mL/kg/hr	Urinary Output ≥ 0.5 mL/kg/hr
> 8	> 12	Administer furosemide and reevaluate in 1 hr	Administer furosemide and reevaluate in 4 hr
4–8	8–12	Administer fluid bolus and reevaluate in 1 hr	Administer furosemide and reevaluate in 4 hr
< 4	< 8	Administer fluid bolus and reevaluate in 1 hr	Monitor and reevaluate in 4 hr

[a]Initial recommended furosemide dosing = 20-mg bolus or 3-mg/hr infusion. Recommended 2-fold incremental increase in subsequent furosemide doses up to a maximum of 160-mg bolus/24-mg/hr infusion rate or until goal CVP/pulmonary artery occlusion pressure achieved. See clinical trial publication for further details.
CVP = central venous pressure; MAP = mean arterial pressure.

d. The RADAR-2 trial also evaluated the conservative and usual care fluid strategies in critically ill patients (n=180) (Intensive Care Med 2022;48:190-200). Although this randomized clinical trial was not exclusively conducted in ARDS patients, subgroup analysis in ARDS was planned a priori. In the ARDS subgroup, no differences in mortality, mechanical ventilation duration, or ICU length of stay were observed between treatment arms. However, it must be noted that the sample size of ARDS patients in the intervention (n=19) and usual care (n=11) study groups was very small, which may limit clinical interpretation for this patient population.

5. NMBAs
   a. Early administration (within 48 hours of ARDS onset) of cisatracurium over 48 hours has consistently improved oxygenation without increasing the risk of neuromuscular weakness.
   b. The ACURASYS trial showed decreased 90-day adjusted mortality with cisatracurium compared with placebo (HR 0.68; 95% CI, 0.48–0.98; p=0.04). Subgroup analysis suggested the greatest mortality benefit in patients with ARDS having a Pa_{O_2}/F_{IO_2} less than 120 mm Hg. Cisatracurium was also associated with significantly more ventilator-free days (up to 28 and 90 days) and organ failure–free days (up to 28 days). Cisatracurium did not significantly increase the risk of ICU-acquired neuromuscular weakness (N Engl J Med 2010;363:1107-16; Crit Care Med 2017;45:446-53).
   c. The ROSE trial showed no significant differences in 90-day mortality between cisatracurium (42.5%) and placebo (42.8%) (between-group difference -0.3 percentage points; 95% CI, -6.4 to 5.9; p=0.93). No differences between study groups were found for secondary end points at day 28, including in-hospital mortality, days free of MV, and days not in the ICU or hospital (N Engl J Med 2019;380:1997-2008).
   d. The optimal strategy for NMBA dosing and monitoring in ARDS remains debatable. The ACURASYS and ROSE trials had similar dosing strategies of a cisatracurium 15-mg bolus followed by continuous infusion at 37.5 mg/hour over 48 hours without titration or train-of-four monitoring. Additional boluses were allowed in patients with elevated plateau pressures (i.e., greater than 30 cm H_2O), given concerns for increased risk of ventilator-induced lung injury.
   e. The 2019 clinical guidelines recommend that early and short-course NMBAs be considered in patients with moderate-severe ARDS (Pa_{O_2}/F_{IO_2} less than 150 mm Hg) (BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69). However, these guidelines were published before the results of the ROSE trial. However, the 2023 clinical practice guidelines recommend against routine use of NMBAs in ARDS (strong recommendation, moderate level of evidence) (Intensive Care Med 2023;49:727-59).

6. Corticosteroids
    a. Proposed rationale for corticosteroid use in ARDS was to prevent fibroproliferation with subsequent alveolar fibrosis from a proinflammatory response.

Table 4. Clinical Practice Guideline Recommendations for Corticosteroids in ARDS

Source	Summary	GRADE Recommendation
Crit Care Med 2017;45:2078-88	Suggest use in early moderate to severe ARDS (Pa_{O_2}/Fi_{O_2} < 200 mm Hg AND within 14 days of onset)	Conditional recommendation; moderate quality of evidence
BMJ Open Respir Res 2019;6:e000420	Limited quality data, resulting in inability to make a recommendation; rather, states that further research is warranted	Research recommendation
Ann Intensive Care 2019;9:69	Not reported	Not reported

    b. Steroids may improve oxygenation, but randomized trials found no definitive mortality benefit. Although the optimal timing of use remains unknown, consideration may be given in patients with early (less than 7 days) and late (7 days or more), whereas initiation beyond 14 days after onset was associated with higher death rates in the LaSRS trial. (N Engl J Med 2006;354:1671-84; JAMA 1998;280:159-65; Crit Care Med 2017;45:2078-88). However, increased death risk in patients initiated on steroids after 14 days of ARDS onset may be misleading. It should be noted this specific cohort had significant baseline characteristic imbalances as well as atypically low death rates in the control group for 60-day (8%) and 180-day (12%) mortality (N Engl J Med 2006;354:1671-84). Furthermore, no significant differences in mortality were found between steroid and control groups in the greater than 14 days cohort after adjusting for important baseline covariates (Crit Care 2007;11:425).
    c. Published guidelines provide minimal direction for clinical practice decision-making regarding the best approach for corticosteroid use in ARDS (BMJ Open Respir Res 2019;6:e000420; Ann Intensive Care 2019;9:69; Crit Care Med 2017;45:2078-88). The 2017 guidelines suggest use in selected patients with ARDS, whereas neither of the 2019 guidelines provides any recommendations because of the paucity of quality data (Table 4).
    d. The DEXA-ARDS clinical trial was the first randomized, placebo-controlled clinical trial investigating dexamethasone for patients with early moderate to severe ARDS (Lancet Respir Med 2020;8:267-76). The corticosteroid group consisted of dexamethasone 20 mg intravenously daily over the first 5 days, followed by 10 mg intravenously daily for up to 10 days or upon extubation (if occurring prior to day 10). Dexamethasone was associated with increased mean ventilator-free days over placebo at 1 month (12.3 ± 9.9 and 7.5 ± 9.9 days, respectively; $p<0.0001$). Other secondary outcomes, including mortality (all-cause, ICU, in-hospital) and MV duration (among all ICU survivors at day 60), significantly favored the dexamethasone group. Of note, hyperglycemia, secondary infections, and barotrauma were similar between study groups. The novelty of this trial compared with previously published corticosteroid studies was the concurrent use of contemporary lung-protective MV strategies. Of importance, this clinical trial was released after publication of the clinical practice guidelines mentioned earlier.
7. Corticosteroids in COVID-19
    a. Currently, clinical practice guidelines strongly recommend the use of dexamethasone in ICU patients with COVID-19 (Table 5).

Table 5. Clinical Practice Guideline Recommendations for Corticosteroids in Critically Ill Patients with COVID-19

Source	Summary	GRADE Recommendation
www.covid19treatmentguidelines.nih.gov Last updated October 10, 2023	Recommends dexamethasone for most patients requiring mechanical ventilation or ECMO	Strong recommendation
www.idsociety.org/COVID19guidelines Last updated September 25, 2020	Recommends dexamethasone rather than no dexamethasone	Strong recommendation, moderate certainty of evidence
Crit Care Med 2021;49:e219-234	• Recommends using a short course of systemic corticosteroids over not using corticosteroids • Suggests specifically using dexamethasone over other corticosteroid derivatives	• Strong recommendation, moderate level of evidence • Weak recommendation, very low level of evidence

b. The landmark RECOVERY clinical trial compared dexamethasone 6 mg daily up to 10 days total administered either orally or intravenously compared to usual care alone in 6425 hospitalized patients with COVID-19 (N Engl J Med 2021;384:693-704). Among 1007 patients in the prespecified subgroup analysis requiring invasive MV, the 28-day morality rate was significantly lower in the dexamethasone group (29.3%) than in the usual care group (41.4%) (RR 0.64; 95% CI, 0.51–0.81). In addition, the probability of hospital discharge within 28 days among those receiving invasive MV significantly improved in patients receiving dexamethasone over usual care (RR 1.45; 95% CI, 1.13–1.85). Moreover, MV cessation was more likely with dexamethasone than with usual care (RR 1.47; 95% CI, 1.20–1.78). Of note, use of dexamethasone lowered overall mortality among all hospitalized patients compared with usual care. However, a survival benefit was only observed among patients with COVID-19 requiring oxygen support (both invasive and noninvasive) over those without any respiratory support.

c. The CoDEX clinical trial randomized 299 adult patients with COVID-19 with moderate to severe ARDS to either dexamethasone (20 mg intravenously daily over 5 days followed by 10 mg intravenously daily for another 5 days or until ICU discharge) or usual care (JAMA 2020;6;324:1307-16). The primary end point was ventilator-free days during the initial 28 days consisting of the number of days alive and liberation from MV for 48 hours or more. The mean number of ventilator-free days in the dexamethasone and usual care groups was 6.6 days and 4.0 days (adjusted difference of 2.26 days; 95% CI, 0.2–4.38; p=0.04), respectively. No differences were observed between study groups for secondary aims, including 28-day mortality, ICU-free days, overall MV days, or risk of adverse events. A limitation of this trial is that it was terminated early because of the RECOVERY trial findings publication. This may have introduced bias despite showing a positive benefit associated with dexamethasone.

d. The WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group conducted a meta-analysis of seven randomized clinical trials involving 1703 critically ill patients with COVID-19. It should be noted most patients included in pooled analysis did not meet a strict definition of ARDS but were considered critically ill. This meta-analysis evaluated the impact of systemic corticosteroids, including dexamethasone, methylprednisolone, or hydrocortisone, on 28-day mortality (JAMA 2020;324:1330-41). Overall pooled data among all trials showed that systemic corticosteroids significantly reduced the risk of all-cause mortality (summary OR 0.66; 95% CI, 0.53–0.82; $p<0.001$) using a fixed-effects model. However, no association was observed between corticosteroids and controls using a random-effects meta-analysis approach (OR 0.70; 95% CI, 0.48–1.01). Subgroup analysis showed the survival benefit favored dexamethasone over no steroids according to pooled data from three clinical trials. However, of note, this benefit with dexamethasone was largely driven by the RECOVERY trial consisting of 57% of the overall pooled data. The remaining clinical trials consisting of hydrocortisone (n=3) and methylprednisolone (n=1) showed that these agents did not affect survival according to subgroup analysis. No increased risk of adverse events was found between corticosteroid and control groups.

e. The optimal dexamethasone dosing strategy remains unknown, though most clinicians have adopted the 6-mg/day approach used in the RECOVERY trial. Of note, higher dexamethasone dosing strategies (20 mg/day) have not shown additional beneficial effects compared with lower doses (6 mg daily) (JAMA 2020;324:1330-41).

8. Inhaled pulmonary vasodilators

a. Inhaled nitric oxide and inhaled epoprostenol have improved gas exchange, despite their lack of effect on clinical outcomes (length of stay and mortality). Limited data suggest these agents are equally efficacious. A meta-analysis of randomized clinical trials comparing inhaled nitric oxide and placebo showed no effect on mortality in patients with ARDS (RR 1.10; 95% CI, 0.94–1.29), patients with a baseline Pa_{O_2}/F_{IO_2} of 100 mm Hg or less (RR 1.01; 95% CI, 0.78–1.32), or patients with a baseline Pa_{O_2}/F_{IO_2} greater than 100 mm Hg (RR 0.89; 95% CI, 0.89–1.42) (Crit Care Med 2014;42:404-12).

b. Institutions have used inhaled epoprostenol as a more cost-effective option over inhaled nitric oxide. Direct comparisons between these studies suggest safety and efficacy were similar. However, inhaled epoprostenol has been associated with significant cost savings over inhaled nitric oxide. Major limitations of widespread inhaled nitric oxide use for ARDS include high costs and dedicated equipment for drug delivery. Given that cost is the primary differentiating factor between these two agents, many institutions have transitioned from inhaled nitric oxide to inhaled epoprostenol. Implementation of inhaled epoprostenol delivery systems and processes requires continuous education and pharmacovigilance to mitigate the risk of medication errors and preventable adverse drug events.

c. The 2019 clinical guidelines did not evaluate inhaled epoprostenol; thus, they do not provide any recommendations on its use in ARDS. However, one of the 2019 guidelines states not to use inhaled epoprostenol in ARDS because of weak quality of evidence (BMJ Open Respir Res 2019;6:e000420), whereas the other provides an expert opinion statement for consideration in severe ARDS for patients currently receiving lung-protective ventilation and proning, but before ECMO (Ann Intensive Care 2019;9:69).

d. Despite the paucity of data for inhaled epoprostenol, it is commonly used in the ICU for ARDS as a bridge to more invasive management strategies (i.e., ECMO) and/or as salvage therapy. Although the optimal dosing regimen remains unknown, the most commonly used doses include a weight-based strategy (typically 10–50 ng/kg/minute titrated to effect) (J Pharm Pract 2019;32:347-60). A small prospective study evaluated a dose-response relationship of weight-based inhaled epoprostenol in patients with ARDS, titrating from 0 to 50 ng/kg/minute by increments of 10 ng/kg/minute every 30 minutes (Chest 2000;117:819-27). A significant increase in the absolute median Pa_{O_2}/F_{IO_2} was observed with the 50 ng/kg/minute weight-based dose compared with baseline (202.2 vs. 187.2 mm Hg, respectively; $p<0.008$). However, no significant difference was observed with the median Pa_{O_2}/F_{IO_2} (183.2 mm Hg) associated with the 10 ng/kg/minute regimen compared with baseline. Furthermore, no statistically significant difference was found between the 10 and 50 ng/kg/minute weight-based regimens regarding oxygenation indices. A major caveat of this study was small sample size (n=9), and the patient population was not reflective of contemporary clinical use of inhaled epoprostenol in moderate to severe ARDS, given that only three patients had a baseline Pa_{O_2}/F_{IO_2} less than 150 mm Hg. Recently, one study observed a similar effect on oxygenation using fixed-dose inhaled epoprostenol without titration compared with inhaled nitric oxide (J Intensive Care Med 2021;36:466-76).

e. One study directly compared weight-based and fixed-dose inhaled epoprostenol in ARDS patients (n=294) (Am J Health System Pharm 2023;80:S11-22). The mean Pa_{O_2}/F_{IO_2} increase from baseline to 4 hours after inhaled epoprostenol initiation was significantly higher in the the fixed-dose group compared with the weight-based dosing approach (81.1 ± 106.0 vs. 41.0 ± 72.5 mm Hg, respectively; p=0.0015). Also, responder rates at 4 hours after inhaled epoprostenol initiation was significantly higher with fixed-dosing over weight-based dosing (69.9% vs. 30.1%; p=0.02). Clinical outcomes were comparable between the two dosing groups.

Patient Cases

1. A 56-year-old man is admitted to the ICU with ARDS after experiencing increasing dyspnea during the past 24 hours. His medical history is significant for alcoholism and hypertension. Results of the initial arterial blood gas are as follows: pH 7.24, Pa_{CO_2} 58 mm Hg, HCO_3 24 mEq/L, Pa_{O_2} 50 mm Hg, and Sa_{O_2} 84% while receiving MV AC mode with F_{IO_2} 100%. Chest radiography reveals diffuse bilateral infiltrates. The patient has blood pressure 120/40 mm Hg (MAP 67 mm Hg), heart rate 142 beats/minute, and CVP 8 mm Hg while receiving a norepinephrine (10 mcg/minute) infusion after intravenous fluid resuscitation. Norepinephrine has now been weaned off while maintaining a MAP of 67 mm Hg. Ceftriaxone 1 g intravenously every 24 hours and levofloxacin 750 mg intravenously every 24 hours have been initiated for community-acquired pneumonia. Which is the best therapeutic plan for the patient's ARDS?
   A. Continue fluid resuscitation to maintain a CVP of 10–14 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; prone positioning; and sedative administration to target deep sedation and cisatracurium infusion.
   B. Begin diuresis to target a CVP less than 4 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; supine positioning; and sedative administration to target deep sedation and cisatracurium infusion.
   C. Begin diuresis to target a CVP less than 4 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; supine positioning; and sedative administration to target deep sedation.
   D. Discontinue fluid resuscitation and begin diuresis to target a CVP less than 4 mm Hg while maintaining a MAP greater than 65 mm Hg; low tidal volume strategy of 4–8 mL/kg of ideal body weight; prone positioning; and sedative administration to target deep sedation and cisatracurium infusion.

Patient Cases *(continued)*

2. A 70-year-old woman (height 63 inches, weight 65 kg) is transferred to your ICU from an outside hospital after outside hospital admission for hypoxic respiratory failure. She had been treated for ARDS at the outside hospital for 3 days before her son requested hospital transfer. On admission, the patient is receiving MV with the following settings: SIMV mode, tidal volume 600 mL (12 mL/kg), respiratory rate 12 breaths/minute, PS 10 cm H_2O, and PEEP 10 cm H_2O. Which is the best therapy for her ARDS?
   A. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; supine positioning.
   B. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; prone positioning; cisatracurium administration.
   C. SIMV mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; supine positioning; cisatracurium administration.
   D. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O; prone positioning.

3. A 63-year-old woman (height 165 cm; total body weight 64 kg) with a medical history of insulin-dependent diabetes and chronic kidney disease was admitted to the general ward 2 days ago for acute COVID-19 pneumonia requiring oxygen therapy via nasal cannula. She was started on dexamethasone 6 mg oral daily. Today, she transferred to the intensive care unit following endotracheal intubation because of worsening oxygenation and meeting severe ARDS criteria. Lung-protective MV is initiated at tidal volume 340 mL, respiratory rate 22 breaths/minute, PEEP 12 cm H_2O, and F_{IO_2} 100%. Her arterial blood gas values are as follows: pH 7.28, P_{aCO_2} 46 mm Hg, P_{aO_2} 60 mm Hg, and HCO_3 24 mEq/L with S_{aO_2} of 92%. Which is most appropriate therapeutic plan regarding her corticosteroids?
   A. Discontinue dexamethasone 6 mg oral daily.
   B. Change dexamethasone 6 mg oral daily to methylprednisolone 40 mg intravenous four times daily.
   C. Change dexamethasone 6 mg oral daily to dexamethasone 6 mg intravenous daily.
   D. Change dexamethasone 6 mg oral daily to dexamethasone 20 mg intravenous daily.

4. K.J. is a 56-year-old man (height 72 inches; total body weight 120 kg) admitted to the ICU for septic shock requiring vasoactive support and ARDS pneumonia. Lung-protective MV is initiated as follows: tidal volume 600 mL, respiratory rate 24 breaths/minute, PEEP 14 cm H_2O, and F_{IO_2} 100% with plateau pressure of 24 mm Hg. His arterial blood gas values are pH 7.28, P_{aCO_2} 46 mm Hg, P_{aO_2} 62 mm Hg, and HCO_3 24 mEq/L. Which of the following treatment strategies may be most appropriate at this time to improve oxygenation?
   A. Inhaled nitric oxide.
   B. ECMO.
   C. Inhaled epoprostenol.
   D. Cistracurium.

5. D.J. is a 49-year-old man with septic shock requiring vasoactive support and ARDS with MV at the following settings: tidal volume 400 mL, respiratory rate 26 breaths/minute, PEEP 20 cm H_2O, and F_{IO_2} 100%, resulting in a plateau pressure of 40 mm Hg. The patient's arterial blood gas values after mechanical ventilation initiation are as follows: pH 7.45, $PaCO_2$ 34 mm Hg, P_{aO_2} 110 mm Hg, and HCO_3 24 mmol/L with S_{aO_2} 88%. Which of the following strategies is most appropriate at this time?
   A. Cisatracurium, furosemide, and prone positioning.
   B. Cistatracurium, furosemide, and ECMO.
   C. Prone positioning, dexamethasone, and cisatracurium.
   D. Prone positioning, dexamethasone, and ECMO.

II. INTUBATION

A. Endotracheal Intubation
   1. Provides access for suctioning of tracheobronchial secretions, maintains a patent airway, and allows administration of medications
   2. Indications include airway protection, facilitation of ventilation and oxygenation, assurance of airway patency, and anesthesia and surgery.
   3. Orotracheal intubation is preferred for elective and emergency cases.
   4. Nasotracheal intubation is beneficial for patients undergoing maxillofacial surgery or dental procedures and for patients with limited mouth opening. May be associated with increased risk of bleeding and sinusitis and should be avoided in patients with severe nasal or midface trauma
   5. Complications include insertion trauma, gastric aspiration, hypoxemia, laryngospasm, esophageal intubation, right main bronchus intubation, cardiac arrhythmias, and hemodynamic impairment

B. Rapid Sequence Intubation (RSI)
   1. Rapid and sequential administration of a rapid-acting induction agent and a NMBA to facilitate intubation and decrease the risk of aspiration
   2. Series of seven distinct steps: preparation, preoxygenation, pretreatment, induction either prior or with paralysis, protection and positioning, placement of the tube in the trachea, and management after intubation

C. Clinical Practice Guideline on Rapid Sequence Intubation
   1. These recently published guidelines were developed to provide evidence-based recommendations on pharmacologic and nonpharmacologic topics related to RSI (Critical Care Medicine 2023;51:1411-30).
   2. Overall, 10 clinically relevant guideline statements involving specific aspects of RSI were finalized; however, a summary of medication-related aspects of RSI are summarized below (Table 6).

Table 6. Clinical Practice Guideline Rapid Sequence Intubation Medication-Related Recommendations

RSI Step	Summary	GRADE Recommendation
Pretreatment	Not reported	Not reported
Induction	Suggest sedative-hypnotic induction agent when neuromuscular blocking agent will be used	Best practice statement (level of evidence not graded)
	No difference suggested between etomidate and other induction agents regarding outcomes (survival, hypotension risk, or need for vasoactive support)	Conditional (moderate level of evidence)
	Suggest against the use of corticosteroids following etomidate administration	Conditional (low level of evidence)
Paralysis	Recommend administering neuromuscular blocking agent following induction	Strong (low level of evidence)
	Suggest administering either rocuronium or succinylcholine (if no contraindications)	Conditional (low level of evidence)

RSI = rapid sequence intubation.

D. Pretreatment
   1. Occurs before an induction agent or NMBA is administered
   2. Pretreatment attenuates the sympathetic and parasympathetic responses (catecholamine release, hypertension, tachycardia, potentially increased ICP in patients with impaired cerebral autoregulation) to laryngoscopy.
   3. Fentanyl or lidocaine can be used as a pretreatment medication (Table 7).
   4. Atropine and defasciculating doses of nondepolarizing NMBAs are not recommended for routine use in RSI for adult patients.

Table 7. Pretreatment Agents

Agent	Dose	Onset	Duration	Advantages	Disadvantages
Fentanyl (Sublimaze)	IV: 1–3 mcg/kg	< 30 s	0.5–1 hr	• Blunts hypertensive response from intubation • Recommended over other opioids because of its rapid onset and short duration of action	• Chest wall rigidity (doses > 100 mcg/kg) • Hypotension, bradycardia, and respiratory depression
Agent	**Dose**	**Onset**	**Duration**	**Advantages**	**Disadvantages**
Lidocaine (Xylocaine)	IV: 1.5 mg/kg	45–90 s	10–20 min	• May prevent increase in ICP through blunting of cough reflex • May reduce bronchospasm in patients with reactive airway disease	• Contraindicated in patients with an amide anesthetic allergy, bradycardia, or severe heart block

ICP = intracranial pressure; IV = intravenous(ly).

E. Induction Agents
   1. Given as rapid intravenous push immediately before the paralyzing agent to help achieve optimal conditions for intubation
   2. Agents should provide rapid loss of consciousness, analgesia, amnesia, and stable hemodynamics.
   3. Agents used for induction during RSI include barbiturates, benzodiazepines (midazolam), etomidate, ketamine, and propofol (Table 8).
   4. Barbiturates
      a. Thiopental is no longer available in the United States.
      b. Methohexital is rarely used because of its adverse effect profile, including respiratory depression, hypotension, and histamine release.
   5. Etomidate
      a. A nonbarbiturate, imidazole derivative with a rapid onset of action and a very short elimination half-life
      b. Enhances the effects of γ-aminobutyric acid, thereby blocking neuroexcitation and inducing unconsciousness (does not provide analgesia)
      c. Transiently inhibits the conversion of cholesterol to cortisol by inhibiting 11ß-hydroxylase, leading to transient adrenal suppression
         i. No convincing or consistent evidence suggests that etomidate is associated with an increased risk of death (Intensive Care Med 2011;37:901-10; Chest 2015;147:335-46).
         ii. Large randomized, prospective, adequately powered studies are needed to clarify the clinical significance of etomidate's effects in patients at risk of adrenal insufficiency.

Table 8. Induction Agents

Agent	Dose	Onset	Duration	Advantages	Disadvantages
Etomidate (Amidate)	IV: 0.2–0.6 mg/kg	10–20 s	4–10 min	• Minimal cardiovascular effects • Decreased ICP with minimal effects on cerebral perfusion	• Myoclonic jerks • Transient decrease in cortisol production
Ketamine (Ketalar)	IV: 1–2 mg/kg IM: 4–10 mg/kg	IV: 10–15 s IM: 3–4 min	IV: 5–15 min IM: 12–25 min	• Catecholamine reuptake inhibition (transient increase in blood pressure and heart rate) • Respiration and airway reflexes maintained • Does not increase ICP • Relieves bronchospasm • Has both amnestic and analgesic effects	• Negative inotropic/chronotropic effects (in catecholamine-depleted patients) • Emergence delirium, nightmares, and hallucinations (premedication with a benzodiazepine does not reduce the incidence)
Midazolam (Versed)	IV or IM: 0.2–0.3 mg/kg	60–90 s	1–4 hr	• Recommended over other benzodiazepines because of its relatively faster onset of action	• Compared with other induction agents, slower onset and longer duration • Dose-dependent respiratory depression and hypotension
Propofol (Diprivan)	IV: 1.5–2.5 mg/kg	15–45 s	3–10 min	• Decreases ICP; however, may also decrease CPP • Mild bronchodilating effects • Drug of choice in pregnancy	• Hypotension and bradycardia • Negative inotropic effects

CPP = cerebral perfusion pressure; IM = intramuscular(ly).

F. NMBAs
   1. Block impulse transmission at the neuromuscular junction, resulting in skeletal muscle paralysis
   2. Used immediately after induction to help achieve optimal conditions for intubation
   3. Have no sedative, analgesic, or amnestic properties
   4. Problematic in patients with a difficult or failed airway
   5. Depolarizing NMBAs (Table 9)
      a. Succinylcholine: Noncompetitively binds to acetylcholine receptors, leading to sustained depolarization of the neuromuscular junction and prevention of muscle contraction
      b. Preferred agent for RSI
   6. Nondepolarizing NMBAs (Table 9)
      a. Rocuronium and vecuronium: Competitive antagonists of acetylcholine at the neuromuscular junction, leading to the prevention of muscle contraction
      b. Intermediate-acting nondepolarizing agents are alternatives when succinylcholine is contraindicated.
      c. Usually have a slower onset of action and longer duration of action
   7. Reversal of nondepolarizing NMBAs with a failed airway
      a. An acetylcholinesterase inhibitor (neostigmine or pyridostigmine) in combination with an anticholinergic (atropine or glycopyrrolate)
      b. Sugammadex (Bridion) binds the aminosteroid class of nondepolarizing NMBAs (vecuronium and rocuronium).

Table 9. Common Neuromuscular Blocking Agents

Agent	Dose	Onset	Duration	Cautions
Succinylcholine (Anectine)	IV: 1–2 mg/kg IM: 3–4 mg/kg (max 150 mg)	IV: 1 min IM: 2–3 min	IV: 3–5 min IM: 10–30 min	• Prolonged effects in pseudocholinesterase deficiency • Hyperkalemia or patients at risk of hyperkalemia (prolonged immobilization, crush injuries, myopathies, burns, muscular dystrophy, stroke, and spinal cord injuries) • Malignant hyperthermia • Bradycardia/hypotension with repeated doses • Mild increase in ICP
Rocuronium (Zemuron)	IV: 0.6–1.2 mg/kg	1–2 min	30–60 min	• Moderate increase in duration with liver dysfunction, minimal increase in duration with renal dysfunction
Vecuronium (Norcuron)	IV: 0.08–0.1 mg/kg	2–3 min	20–60 min	• Prolonged duration in renal and liver dysfunction

G. Management After Intubation
   1. Provide continued sedation/analgesia as needed if an intermediate-acting NMBA was used and assists in adequate oxygenation and ventilation
   2. Minimize long-term use of analgesics and sedatives
   3. Maintain head-of-bed elevation at 30–45 degrees
   4. Mouth and eye care
   5. Bowel regimen
   6. Stress ulcer and deep venous thrombosis prophylaxis

Patient Cases

6. A 55-year-old man (weight 75 kg) is admitted to the burn ICU after a 65% total body surface area burn to the abdomen, back, and lower extremities from a house fire. He is unconscious and unable to protect his airway. His medical history is significant for hypertension and hyperlipidemia. He is currently receiving high-dose norepinephrine and vasopressin to maintain a MAP of 65 mm Hg. His current laboratory test results show the following: sodium (Na) 130 mEq/L, potassium (K) 5.9 mEq/L, chloride (Cl) 122 mEq/L, carbon dioxide (CO_2) 15 mg/dL, blood urea nitrogen (BUN) 10 mg/dL, and serum creatinine (SCr) 1.3 mg/dL. Which group of medications would be most appropriate for RSI?
   E. Fentanyl, propofol, rocuronium.
   F. Fentanyl, ketamine, succinylcholine.
   G. Fentanyl, etomidate, rocuronium.
   H. Fentanyl, propofol, succinylcholine.

7. A 39-year-old homeless man (weight 70 kg) was admitted to the neurosciences ICU with a traumatic head injury after falling off a 3-ft ladder while intoxicated. Imaging reveals a subdural hematoma. The team decides to intubate this patient. His current laboratory values are as follows: Na 133 mEq/L, K 4.5 mEq/L, Cl 97 mEq/L, CO_2 28 mg/dL, BUN 13 mg/dL, SCr 0.7 mg/dL, and glucose 140 mg/dL. Which induction medication would be most appropriate to use for RSI?
   I. Propofol 90 mg intravenous push.
   J. Ketamine 100 mg intravenous push.
   K. Midazolam 15 mg intravenous push.
   L. Etomidate 150 mg intravenous push.

MECHANICAL VENTILATION

A. Critical to Understanding How MV Works: A fundamental knowledge of acid-base disorders and normal respiratory physiology

B. Two Essential Categories of Respiratory Failure: Hypercapnic and hypoxemic. Derangements in Pa_{O_2} or Pa_{CO_2} will help determine the cause of respiratory failure. (Table 10 provides the context for normal oxygenation and ventilation values.)

C. Modes
   1. Assist control (AC) ventilation
      a. Volume control (VC)
         i. The patient receives a predetermined respiratory rate and tidal volume, with additional patient-initiated breaths provided at the preset tidal volume. Patient-initiated respiration generates a negative pressure within the ventilator circuit, which is sensed by the ventilator, and a full tidal volume breath is provided.
         ii. Potential for ventilator dyssynchrony, "double-stacking," and respiratory alkalosis
         iii. Mode used in the ARDSNet trial of tidal volume strategy to limit spontaneous tidal volumes (N Engl J Med 2000;342:1301-8)
      b. Pressure control
         i. The patient will receive a breath at a fixed rate until a predetermined peak pressure limit is reached. The tidal volume is variable and limited by the peak pressure limit.
         ii. Not ideal for patients with low minute ventilation and may lead to hypoventilation and further hypoxia
   2. Synchronized intermittent mandatory ventilation (SIMV): Patients will receive a predetermined respiratory rate and tidal volume plus additional spontaneous, self-generated breaths at whatever tidal volume they can generate. Not ideal for the treatment of ARDS, given patients' ability to exceed the present tidal volume for spontaneous breaths in excess of 6 mL/kg
   3. PS ventilation
      a. Usually used as a weaning mode of MV from a more intensive mode of MV (i.e., AC ventilation)
      b. The patient initiates each breath with assistance from the ventilator in the form of a preset pressure value. The ventilator is set to provide a correct amount of pressure to assist with each inspiratory effort. The tidal volume and respiratory rate depend on the patient.

D. Ventilator Variables
   1. F_{IO_2}
      a. Amount of oxygen that is delivered with each breath, from 21% to 100% at sea level
      b. F_{IO_2} is generally tapered to provide the minimal oxygenation needed to meet patient needs. Concerns for oxygen toxicity with high F_{IO_2} requirements over prolonged period
   2. Tidal volume
      a. Volume of air inspired in a breath (delivered by MV or spontaneously)
      b. Set according to oxygenation and ventilation requirements. Patients with ARDS are treated with a low tidal volume strategy.
   3. Respiratory rate
      a. Set to provide a minimal number of breaths from the ventilator at the set tidal volume
      b. Titrated by minute ventilation, Pa_{CO_2}, and pH. Minute ventilation (liters per minute) = tidal volume (liters) × respiratory rate (breaths/minute).

4. Flow rate
   a. Velocity of air delivered
   b. Velocity is greatest initially upon inspiration and decelerates toward the end of the inspiratory effort.
5. PEEP
   a. Positive pressure in the alveoli during expiration
   b. Pressure at the end of expiration promotes alveoli recruitment while minimizing the risk of collapse in ARDS.
   c. Titrated with F_{IO_2} to provide the minimum necessary support to meet the patient's oxygenation requirements

Table 10. Normal Respiratory Physiology Values

Value	Normal Range
Tidal volume (mL/kg)	5–10
Respiratory rate (breaths/min)	12–20
Minute ventilation (L/min)	5–10
Pa_{CO_2} (mm Hg)	35–45
Pa_{O_2} (mm Hg)	80–100
Sa_{O_2} (%)	95–100

Patient Case

8. A 74-year-old woman (height 63 inches, weight 65 kg) is transferred to your ICU from an outside hospital after admission for hypoxic respiratory failure. She had been treated at the outside hospital for ARDS for 3 days before her son requested transfer. When the patient arrives in your ICU, she is receiving MV with the following settings: SIMV mode, tidal volume 600 mL (12 mL/kg), respiratory rate 12 breaths/minute, PS 10 cm H_2O, and PEEP 10 cm H_2O. Which is the best ventilator plan for treating her ARDS?

   M. AC/VC mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O.
   N. SIMV mode, tidal volume 300 mL (6 mL/kg), respiratory rate 20 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O.
   O. PS mode, PS 10 cm H_2O, PEEP 5 cm H_2O.
   P. SIMV mode, tidal volume 300 mL (6 mL/kg), respiratory rate 10 breaths/minute, PS 10 cm H_2O, PEEP 5 cm H_2O.

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Mechanical Ventilation

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

The patient has early severe ARDS (for less than 48 hours and Pa_{O_2}/F_{IO_2} less than 150 mm Hg) and hemodynamic stability (post-resuscitation MAP greater than 65 mm Hg). According to the findings of an ARDSNet-sponsored multicenter trial, he would qualify for conservative fluid management (CVP less than 4 mm Hg). In addition, vasopressors should be discontinued and diuresis initiated to achieve a target CVP less than 4 mm Hg (Answer A is incorrect). Given the timing and the severity of the patient's ARDS, he should be placed in the prone position (Answers B and C are incorrect). In addition, this patient would qualify for a cisatracurium infusion. Given the timing and severity of his ARDS, he qualifies to receive a lung-protective ventilation strategy (tidal volume 4–6 mL/kg of ideal body weight) and diuresis to a CVP less than 4 mm Hg (hemodynamically stable if weaned off vasopressors) while placed in the prone position and administered a cisatracurium infusion (Answer D is correct).

2. **Answer: A**

The patient presents to your ICU after 3 days of care. Therefore, she does not currently meet the criteria for being administered cisatracurium or placed in the prone position (Answers B–D are incorrect). Currently, the applicable therapy to apply is a lung-protective ventilation strategy (tidal volume 4–6 mL/kg of ideal body weight) (Answer A is correct).

3. **Answer: C**

The patient was appropriately started on corticosteroids for COVID-19 pneumonia during the same hospital admission prior to developing ARDS and subsequently transferred to the ICU. Dexamethasone was associated with favorable results among COVID-19 hospitalized patient requiring oxygen support (both noninvasive and invasive mechanical ventilation) as well as critically ill patients with moderate to severe ARDS; therefore, continuing dexamethasone is appropriate (Answer A is incorrect). Methylprednisolone has not been associated with favorable clinical outcomes among COVID-19 or moderate to severe ARDS patients (Answer B is incorrect). Although the DEXA-ARDS trial found improved clinical outcomes associated with the dexamethasone 20 mg daily (followed by 10 mg daily) regimen in ARDS patients, it is important to recognize this patient population did not consist of COVID-19. Furthermore, a meta-analysis evaluating various corticosteroid regimens failed to find any benefit of higher dexamethasone dosing strategies (i.e., 20 mg daily) compared with lower doses (6 mg daily) (Answer C is correct; Answer D is incorrect).

4. **Answer: B**

Although selective pulmonary vasodilating agents may modestly improve oxygenation parameters at best, they have not shown to improve clinical outcomes. Their role in therapy is uncertain, but it may be mostly considered salvage therapy after failing more proven strategies (Answers A and C are incorrect). The role of neuromuscular blocking agents in ARDS remain controversial. However, the ROSE clinical trial failed to corroborate any improved clinical outcomes previously demonstrated (ACURASYS trial). Cisatracurium may not be appropriate at this time since plateau pressures remain acceptable despite using high PEEP (Answer D is incorrect). ECMO use in ARDS also remains controversial. However, severe ARDS may be a compelling reason to initiate ECMO compared with alternative supportive therapies (Answer B is correct).

5. **Answer: C**

ECMO may not be the most appropriate strategy at this time considering the patient has moderate to severe ARDS with corresponding Pa_{O_2} of 110 mm Hg on lung protective mechanical ventilation alone (Answers B and D are incorrect). Furosemide may not be the most appropriate strategy because the patient is currently requiring vasoactive support (Answers A and B and incorrect). Cisatracurium may be indicated for this patient considering the plateau pressure is elevated as well as prone positioning to help improve oxygenation with suboptimal parameters on mechanical ventilation along; furthermore, corticosteroids may also be indicated given ARDS severity and timing (Answer C is correct).

6. **Answer: C**

Propofol, ketamine, and etomidate can all be used for induction. Propofol may worsen hypotension; therefore, because this patient is already receiving vasoactive medications to maintain his blood pressure,

propofol would not be ideal (Answer A is incorrect). Succinylcholine causes the up-regulation of acetylcholine receptors, predisposing the muscle fibers to release excess potassium because they are depolarized, which leads to significant dysrhythmias or cardiac arrest (Answers B and D are incorrect). Appropriate induction and neuromuscular blockade in this patient would be to administer etomidate (indicated for hemodynamically unstable patients), fentanyl (providing adequate analgesia), and rocuronium (does not increase serum potassium) (Answer C is correct).

7. **Answer: B**

Although propofol may promptly lower ICP, it can also induce hypotension and thus decrease cerebral perfusion pressure (Answer A is incorrect). Midazolam could be used as an induction agent; however, it is not the best agent for RSI because of its delayed onset of action (Answer C is incorrect). Etomidate may be considered in the setting of increased ICP; however, the dose in this case is too high (Answer D is incorrect). Ketamine not only decreases ICP but also prevents fluctuations in ICP (Answer B is correct).

8. **Answer: A**

After recognizing that the patient has ARDS, a lung-protective ventilation strategy should be implemented (tidal volume 4–6 mL/kg). Choosing a PS or SIMV mode would allow the patient to initiate spontaneous breaths in excess of the goal tidal volume (Answers B–D are incorrect). In the ARDSNet study of tidal volume strategy, the AC mode was most commonly used to promote the application of low tidal volumes (Answer A is correct).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: B**

According to the Berlin ARDS definition, the category of acute lung injury was removed in favor of categorizing the severity of ARDS (PaO_2/FIO_2 less than 200 mm Hg) (Answer A is incorrect). Because of the relative difference in mortality rates, mild and moderate ARDS cases are less likely to benefit from therapeutic interventions, given the number needed to treat to show an effective intervention (Answers C and D are incorrect). In the trials evaluating prone positioning and cisatracurium, patients with severe ARDS were the most likely to benefit. Although the criteria used for severe ARDS in these studies differed from those used in the Berlin definition (both studies were initiated before publication of the Berlin ARDS definition), a post hoc analysis shows a survival benefit in favor of the group with the highest mortality rate (Answer B is correct).

2. **Answer: C**

The Belin definition defines ARDS severity with PaO_2/FIO_2 as one element of criteria. Comprehension of subpopulations of ARDS is essential to apply published clinical data among various therapies. Although not universal defined, the most commonly reported inclusion criteria among landmark ARDS clinical trials (ROSE, ACURASYS, DEXA-ARDS, etc.) was moderate to severe ARDS criteria was less than 150 mm Hg (Answer C is correct; Answers A and B are incorrect). Also, these trials did not define a lower limit of ARDS severity (e.g., 50–150 mm Hg) as inclusion criteria (Answer D is incorrect).

3. **Answer: B**

The landmark ARDSNet trial evaluating lung protective mechanical ventilation strategies defined low tidal volumes ranging from 4 to 8 mL/kg using predicted body weight (also known as ideal body weight (Answer B is correct). Total body weight may overestimate ventilation requirements in adult patients with obesity because lung volume is unrelated to increasing total body weight (i.e., lung size remains relatively constant irrespective of adult body mass) (Answers A, C, and D are incorrect).

4. **Answer: B**

This patient has both ARDS and septic shock. In addition, he has likely had ARDS for less than 48 hours; therefore, the time to initiation of several treatments is essential. The patient is actively in shock (as evidenced by his blood pressure), thus making a fluid-conservative strategy (CVP less than 4 mm Hg) impossible (Answer C is incorrect). Because the time to presentation is less than 48 hours and the patient has severe ARDS, he meets the criteria for cisatracurium administration and prone positioning, and a treatment plan should include these two therapies (Answers A and D are incorrect). A therapy plan should include shock resuscitation (fluid-liberal strategy, CVP 10–14 mm Hg), lung-protective ventilation (tidal volume 4–8 mL/kg of ideal body weight), prone positioning, and cisatracurium administration (Answer B is correct).

5. **Answer: D**

The FACTT Lite approach attempts to simplify fluid management for implementation among ARDS patients in clinical practice. The parameters used to determine clinical course of action (remove fluid from diuretic administration, administer fluid boluses, or monitor without administration of either therapy) include MAP, urine output, and CVP (pulmonary arterial occlusion pressure may be optional based on institutional practices for invasive monitoring). Administering fluids in this patient may not be the most appropriate approach because MAP, CVP, and urine output do not suggest hypovolemia (Answer C is incorrect). Although continuous infusion furosemide may be an option within this published protocol, it is important to note both the suggested rate of infusion and time to re-evaluate and inconsistent with the FACTT Lite protocol. Re-evaluation should be relatively short irrespective of the clinical decision-making (i.e., 1 or 4 hours) (Answer B is incorrect). It is important to emphasize conservative management approaches defined in this protocol require the patient achieve a MAP of 60 mm Hg without any vasoactive support for at least 12 hours. The short time period since vasopressor support cessation may be too aggressive to administer without demonstrating clinical stable hemodynamic pressures (Answer A is incorrect). The most appropriate approach would be monitoring until adequate MAP has been achieved without vasoactive support for at least 12 hours (Answer D is correct).

6. **Answer: A**

The DEXA-ARDS clinical trial has been the only contemporary trial evaluating corticosteroids (dexamethasone) among ARDS patients already receiving lung-protective mechanical ventilation strategies (Answer A is correct). Despite the numerous ARDS trials evaluating methylprednisolone, none have concurrently evaluated with lung-protective strategies among all study patients (Answer C is correct). Despite the optimal timing of corticosteroid use in ARDS remains controversial, dexamethasone was started "early" after ARDS onset in the DEXA-ARDS trial (Answer B is incorrect). The LaSRS clinical trial found increased risk of death in the steroid group compared with controls among patients enrolled greater than 14 days after ARDS onset. Unfortunately, this unadjusted subgroup analysis may have biased these findings. Subsequent re-evaluation of these results found no differences in mortality among in this subgroup. Furthermore, "late" (greater than 5–7 days) corticosteroid use in ARDS has not consistently demonstrated worse outcomes (Answer D is incorrect).

7. **Answer: D**

Neuromuscular blocking agents should always be administered after induction agents (Answers A, B, and C are incorrect). In addition, atropine is not routinely recommended (Answer C is incorrect) for RSI in adult patients. Atropine should be kept nearby for patients who are at an increased risk of bradycardia during RSI (use of β-blockers, calcium channel blockers, digoxin, or amiodarone). Induction agents (and pretreatment medications) should be administered before NMBAs (Answer D is correct).

8. **Answer: B**

The patient has moderate to severe ARDS. The initial MV strategy is using low tidal volumes (Answers A and C are incorrect). Considering the patient is hemodynamically stable without requiring vasoactive support, conservative fluid management approaches are warranted (Answer B is correct). Hemodynamically unstable patients may be candidates for liberal fluid management strategies (Answer D is incorrect).